



INSTALLATION, OPERATIONS AND MAINTENANCE MANUAL for

VC / HVC-3500 Pressure and Thermal Vacuum Controller

Revision: A16



LACO Technologies, Inc.
3085 West Directors Row
Salt Lake City, Utah, 84104
Phone: (801) 486-1004 or (800) 465-1004
Fax: (801) 486-1007
www.lacotech.com
info@lacotech.com

Table of Contents

1. Introduction	4
1.1. Contact Information	4
1.2. About the Manual	4
1.3. Warning Messages	5
2. Safety	6
2.1. Intended Use	6
2.2. Owner Requirements	6
2.2.1. Safe Operation	6
2.2.2. Personnel Qualifications	6
2.2.3. Operator Requirements	6
2.3. Dangers	7
2.3.1. Dangers from Electrical Power	7
2.4. Safety Labeling	8
2.5. Equipment Safety Practices	8
2.5.1. Process or System Faults	8
2.5.2. Overtemperature Controller	8
3. Scope of Delivery, Transport, Storage	9
3.1. Shipment Items	9
3.2. Transport & Location.....	9
3.3. Storage and Usage conditions.....	9
3.4. Decommissioning	9
4. Operating Procedures	10
4.1. Switching on and off the device.....	10
4.2. Hardware Interface.....	10
4.3. Software Screens and Parameter Settings.....	10
4.3.1. Info & Startup Screen	10
4.3.2. Home Screen.....	11
4.3.3. Password Screen	12
4.3.4. Advanced Settings Screens.....	14
4.3.4.1. Vacuum Chamber Settings	15
4.3.4.2. System Thermal Settings and PID Parameters	16
4.3.4.3. User Configured Warnings and Faults:.....	19
4.3.5. Maintenance Screens	20
4.3.5.1. IO Maintenance.....	21
4.3.5.2. ASCII Communications Maintenance	22
4.3.5.3. Thermal PID Algorithm.....	23
4.3.5.4. System Pressures, shows the chamber pressure in all the units.....	24
4.3.5.5. System Warning/Fault Status.....	24
4.3.5.6. Recipe Cycle Status	25
4.3.5.7. Temperature Input Calibration.....	25
4.3.5.8. System Logs.....	26
4.3.6. Data Log Screen.....	27
4.3.7. Manual Mode	29
4.3.7.1. Manual Mode – Thermal Manual Screen	30

4.3.7.2. Recipe Screen.....	31
4.3.7.3. Recipe – Vacuum Setpoints	33
4.3.8. Run Screen and Running processes.	33
4.3.9. Trending Graphs	34
5. Remote Connection	36
5.1.1. Connect your computer to the VC/HVC3500 Controller	36
5.1.2. Controlling the VC/HVC3500 remotely	36
5.1.3. Remote accessing files on the SD card in the VC/HVC3500 controller.	36
6. Advanced Unitronics Settings and Downloading System Updates.....	37
6.1. Set the Date and Time.....	37
6.2. Software Update.....	38
6.3. Firmware Update	39
6.4. IP Addresses.....	40
6.4.1. Changing the IP Address	40
6.4.2. Change IP Address for Third Party Devices	42
7. Faults, Warnings and Troubleshooting	44
8. Maintenance and Spare Parts	48
9. Appendix	49
9.1. Communications	49
9.1.1. Commands and Examples.....	49

1. Introduction

1.1. Contact Information

For technical assistance or to order replacement parts call, fax or email from the information listed below.

LACO Technologies
3085 West Directors
Salt Lake City, Utah 84104
www.lacotech.com

Phone: (800) 465-1004
Phone: (801) 486-1004
FAX: (801) 486-1007
info@lacotech.com

Returned equipment will not be accepted without prior authorization. Prior to shipping please call for a returned material authorization (RMA) number. LACO Technologies reserves the right to cancel the warranty if the controller is disassembled without authorization.

1.2. About the Manual

This purpose of this manual is to provide instructions for the installation, operation, and troubleshooting of the **VC / HVC-3500 Controller**. The manual contents will be provided in hardcopy format and available for download at www.lacotech.com. This manual applies to software releases 4.00 or greater. In most cases, systems with a 3500 controller will ship with two manuals. The second manual, called the 3500 system manual, discusses items outside of the scope of the controller itself.

1.3. Warning Messages

See below for the range of warning messages used throughout this manual. The lower text in these messages indicates the severity range of the message group.



DANGER

Imminent threat of death resulting in death or severe injuries.



WARNING

Dangerous situation resulting in major injuries.



CAUTION

Dangerous situation resulting in minor injuries.

NOTICE

Dangerous situation resulting in damage to the equipment, property or environment.

2. Safety

2.1. Intended Use

The 3500 controller is designed to control pressure and thermal processes.

- You must install, operate and service this device only in compliance with these operating instructions.
- Adhere to the storage and usage recommendations.

2.2. Owner Requirements

2.2.1. Safe Operation

- Operate the device only when it is in technically perfect working order.
- Operate the device only as specified in a safety-conscious and hazard-conscious manner and in compliance with these operating instructions.
- Comply with the following regulations and monitor their compliance.
 - Intended use
 - Safety and accident prevention regulations
 - International, national and local standards and guidelines
 - Additional provisions and regulations that are specific to the unit **Safety conscious**
- Use only original parts or parts approved by the manufacturer.
- Keep these operating instructions available at the equipment location.

2.2.2. Personnel Qualifications

- Allow only qualified service technicians to work with and on the device. The qualified service technicians must have received training on the device.
- Allow personnel in training to work with and on the device only under the supervision of trained qualified service technicians.
- Make sure that the authorized personnel have read and understood the operating instructions and all other applicable documents, especially the information regarding safety, maintenance and repairs, before starting work.
- Never perform maintenance on the system with the power on.

2.2.3. Operator Requirements

- Read, observe and follow the information in these operating instructions and the working instructions created by the owner, especially the safety instructions and warnings.
- Carry out any work only based on the complete operating instructions.
- If you have any questions regarding operation or maintenance that you cannot find answers to in these instructions, then please contact LACO customer service.
- When using this equipment, be sure to follow the safety procedures outlined by your facility. These safety procedures should cover the two primary types of hazard training: (1) equipment hazards and (2) facility-related hazards.

- Never override or bypass any hardware or software interlocks.
- All safeguard devices must be in place when equipment is in operation. Operators, set-up operators, helpers or installation personnel should not alter, remove or disable safety equipment.

2.3. Dangers

The device was built according to recognized safety regulations. Nevertheless, improper use or maintenance can result in danger to life and limb of the operator or other persons and damage to the device and other property. In order to prevent safety issues, follow the items listed below.

- Use the equipment only as intended ([section 2.1](#))
- Only authorized and trained personnel should use the equipment. ([section 2.2](#), [2.3](#), [2.4](#))
- Follow all safety labels ([section 1.3](#))
- Follow all safety procedures when servicing equipment.

Below are summarized potential risks or hazards to service or operator personnel.

2.3.1. Dangers from Electrical Power

- Considerable voltages arise inside the device. Touching parts where electrical voltage is applied can result in death.
- Disconnect the device from the electrical power supply prior to any installation and maintenance work. Make sure that the electric power supply is reconnected with authorization. Testing live parts can result in danger to life.
- The device contains electric components that can be damaged from high electric voltage.
- Before connecting the device to the power supply, make sure that the supply voltage specified on the device is the same as the local power supply.
- Keep all liquids away from electrical panels, and connections.

2.4. Safety Labeling

Hazardous Voltages. Identify enclosures with voltages over 24VDC that should be serviced only with the power disconnected.



2.5. Equipment Safety Practices

2.5.1. Process or System Faults

Process or System faults are monitored and controlled by the logic controller (PLC). For more information on Fault troubleshooting see section 7. When a fault occurs the system stops and the screen indicates the specific fault reason. Once the fault is fixed the system will auto recover to the system status state “ready to test”.

!CAUTION

Process or system faults do not remove hazardous voltage from the system. Only authorized, qualified, trained personnel should service this equipment.

2.5.2. Overtemperature Controller

HVC-3500 versions of the 3500 controller that use electric heaters will include a Watlow overtemperature controller. This safety device ensures the heating system turns off in case the heater temperature becomes too hot. Verify the overtemperature controller setpoint is appropriate for the desired max temperature of the system. Typically, the overtemperature setpoint should be set around 30-50 C higher than the maximum system temperature. Controllers that use thermal control units (TCU/chillers) do not have this over temperature controller.

3. Scope of Delivery, Transport, Storage

3.1. Shipment Items

- 3500 Controller
- 3500 Controller hardcopy manual

3.2. Transport & Location

The 3500 controller will be transported in a shipping box or crate with packaging. Remove the 3500 Controller (look for dented panels, paint scratches, etc.). Any damage to the system should be reported to the carrier immediately. Locate the unit with easy access to required utilities (see section 5).

3.3. Storage and Usage conditions

Store and operate the 3500 Controller per the table located in [section 4.3.1](#).

3.4. Decommissioning

The system can be disposed of by the user or sent to LACO Technologies. Please comply with environmental and safety regulations of your country when disposing of the device. The device is made of recyclable materials. This option may be used to avoid waste and protect the environment.

4. Operating Procedures

4.1. Switching on and off the device

Refer to [section 5.2](#) to start the device and [section 5.3](#) to turn off the device.

4.2. Hardware Interface

On the side of the VC-3500 enclosure This switch turns off all power to the machine. The main enclosure cannot be opened while this switch is in the ON position.

On systems that have needle valves or any other kind of hardware that can prevent the chamber from venting when the vent valve is open, be sure to set the local atmospheric pressure in manual mode before you begin operating.

4.3. Software Screens and Parameter Settings

4.3.1. Info & Startup Screen



This screen is shown during system startup or when the info button is pressed from the home screen. This screen indicates the current software revision.

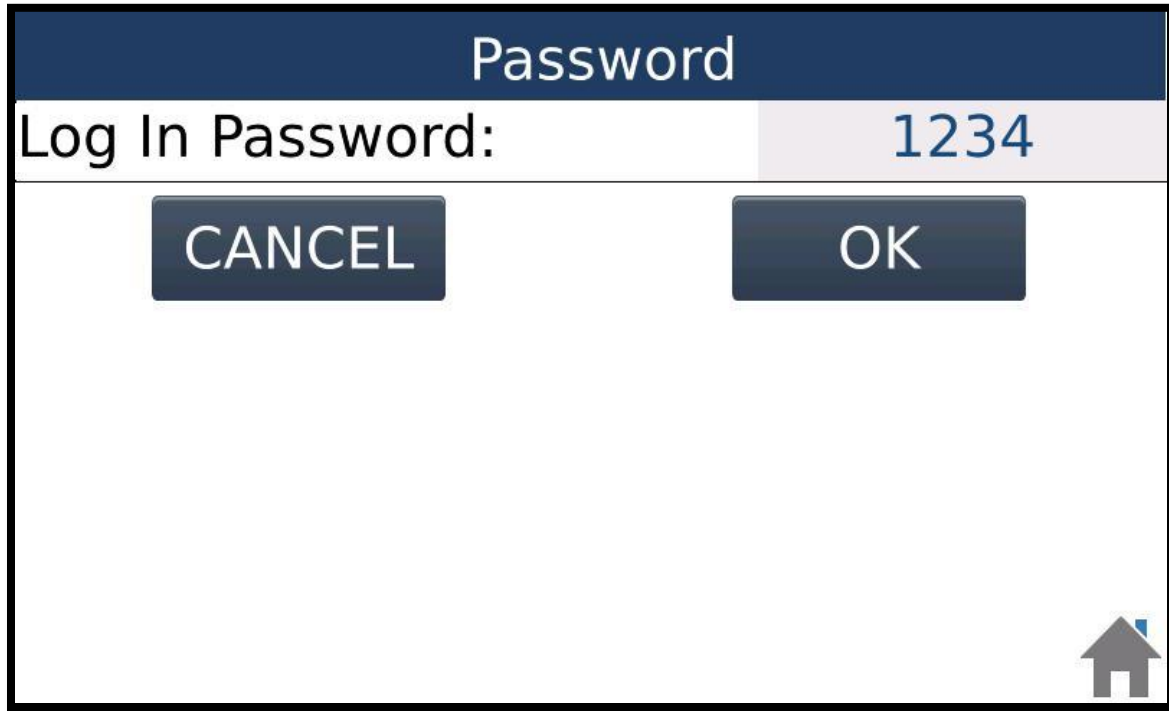
4.3.2. Home Screen



The home screen allows access to all 3500 controller software features. Press the desired icon to access the needed menu. The Home screen is accessible from any screen by pressing on the Home icon in the lower right screen corner.

The user can switch between automode and manual mode from this screen using the toggle switch in the bottom center. Also, pressing the manual icon will switch the system into manual mode, and pressing the Run button will switch the system into auto mode. Depending on the current access level ([see section 6.3.4](#)) some screens icons may be “grayed” out, or inaccessible, due to the current access level. For example, the grayed out icons in the screenshot below are inaccessible due the access level being the default operator level.

4.3.3. Password Screen



The above screen appears by pressing the Log In key icon. Enter the password for desired level of access. There are three access levels on the 3500 controller.

- 1) Operator Level. The Operator Level requires no password. This is the default level when the controller is powered on. In this mode the user can do the following.
 - run tests
 - load but not edit recipes.
 - view system information
 - log in.
- 2) Technician Level. To access the Technician level, press the gray number area above the OK button and enter the password and press OK button. The as-shipped password is 1234. If a correct password is entered the screen will go back to the home menu. If a user is logged in as Technician, they may change the password by entering a 4-digit password twice and pressing the CHANGE button (see screenshot below). The technician login will log out automatically after 15 minutes unless the machine is operating in manual mode. Press the CANCEL button while in Technician Level to manually return to Operator level. The Technician Level gives access to:
 - Edit recipes
 - Edit data logging settings
 - View Maintenance settings
 - Change password
 - Operate the system in Manual Mode

- 3) Advanced Level. To access the Advanced level, enter the advanced password and press OK. The hard coded password is 5226. If a correct password is entered the screen will go back to the home menu. If a user is logged in as Advanced, they may change the password by entering a 4-digit password twice and pressing the CHANGE button (see screenshot below). The advanced login will log out automatically after 15 minutes unless the machine is operating in manual mode. Press the CANCEL button while in advanced level to manually return to Operator Level. The Advanced Level gives access to:
- All the features of the Technical Level
 - Advanced Settings

Password	
Log In Password:	5226
CANCEL	OK
Change Password	
Confirm Password:	0
Enter New Password:	0
	

4.3.4. Advanced Settings Screens

The 3500 controller is a highly configurable controller to meet the demands of customized systems. The advanced settings screens are used to configure the controller for a specific system. It is strongly recommended not to adjust these settings without first consulting with Laco.

⚠ CAUTION

The 3500 controller is shipped with the correct advanced settings. Advanced settings should not be changed, unless consulting with Laco. Use extreme caution when changing advanced settings, because most settings reflect the physical system configuration. Inadvertent changing of advanced settings can make the 3500 system inoperable or a cause unsafe conditions.

NOTICE

All current advanced settings will be identified in the appendix of the manual with a separate signed sheet of paper.

Advanced Settings Menu

Vacuum Settings	Settings	Load Factory Settings	Load Factory
Temperature Control	Settings	Load User Settings	Load User
System Faults	Settings	Save User Settings	Save User
Log In Time	00:10	Always Logged In	☐
Y-axis Graphing Options		Y-axis Temperature Scale	
Scale: Linear		Min	Max
Min	Max	0.000	0.000
0.000000	0.000		

Each settings button on the left side will take you to a configuration screen specific to the category header on the left.

The Log in Time is the time duration in which the system will remain logged in after the user has entered the password. If the user wants the system to always be logged in, then select the “Always Logged In” box.

The right side of the screen has three options:

- Load Factory Settings
- Load User Settings
- Save User Settings

- Load Messages (should only need to be pressed once, unless internal memory is somehow lost).

Before the unit leaves Laco, the factory settings are saved onto the PLC. If at any time the user changes the configurations to an undesirable/unusable state, and is unable to select the correct settings to return to normal, pressing the “Load Factory Settings” button will return all settings to the way they were when the system left the factory.

If at any time the user finds more desirable settings and wants to save those settings for future use, push the “Save User Settings” button. Then if the settings are changed to an undesirable configuration, the user can then press “Load User Settings” to return to the settings at the last saved configuration.

Graph options: This is for the trending graphs used in manual and auto mode:

- Scale: Options are linear or logarithmic
- Min/Max: These are the minimum and maximum values displayed for the respective axes.
 - Note, the trending data in the graph does not update if the scaling is changed. If the scaling is changed, please reset the trending data.

4.3.4.1. Vacuum Chamber Settings

Advanced Pressure Settings	
Chamber Pressure Unit	Torr
Chamber Time Unit	sec
Pump Power	Always On
Pressure Rate Sampling:	00:03
Enable Ascii in Recipe	☐
Use Fill Valve on Reset	☐
High Vac Xover Setpoint	0.010
Enable Recipe Edit in Cycle	☐




- Vacuum Chamber Units Selections – This option dictates which unit you would like to display for pressure. Note that three units represents altitude as it correlates with the pressure inside the chamber. Press the icon and select from the following pressure unit options:

Torr	PSIA	PSIG	inHgA	inHg
Mbar	Pa	Atm	hPa	Feet
Meters	Miles	kPa	Miles	

- Clarification on the inches of mercury units:

- inHgA – absolute pressure, this will read 29.92 at one atmosphere and will approach zero as you approach full vacuum.
- inHg – gauge pressure, this will read zero at the local atmospheric pressure and will increase as you vacuum the chamber down. This unit is a gauge pressure and will only reach 29.92 if the local atmospheric pressure is one atmosphere (sea level). If your chamber is at higher elevations it will not reach 29.92 inHg. For example, in Salt Lake City, this unit will not get higher than 25.8 inHg.
- Pump Power – Select from the following Vacuum pump Power options (only applicable when not in manual mode).
 - Always On – Evac pump is always on. (Mandatory for systems with a turbo pump)
 - On Demand – Evac pump is only on when evac valve is open.
 - On during cycle – Evac pump is on only during test process.
- Enable Ascii in Recipe: This will enable the option in the recipes to allow the user to control the chamber pressure and temperature via the RS232/Ethernet ports using ASCII codes (see appendix).
- Use Fill Valve on Reset: If a fill valve is on the system and the user does not want to use the vent valve while resetting the system to Ambient, select this button and the fill valve will be used instead.

4.3.4.2. System Thermal Settings and PID Parameters

Advanced Thermal Settings

Temp Unit	C	PID Cntrl	PID time	Max. SP	Min. SP	Max Source	Min Source
Max Vent Temp	60.0	Zone 1 Platen	01:00	125.0	-50.0	150.0	-60.0
Min Vent Temp	10.0						

P-GAIN	I-GAIN	D-GAIN	bias-PWM	bias-SP	windup
1.00	0.50	1.00	55.0	50.0	0.5

Buttons: Zone 1, Chiller Setpoint, Sensor Labels, Home icon.

The thermal screen is divided into three portions:

1. Characteristics for the entire chamber
2. Characteristics for each zone in the chamber (most chambers will only have one zone).

3. PID parameters for each thermal source in each zone.

Characteristics for the entire chamber:

- Temperature Unit – Select between degrees Celsius and degrees Fahrenheit
- Max Vent Temp – The maximum allowable temperature in which the chamber can be vented. Reasons for this can range from not wanting to accidentally burn the user if its too hot (if the chamber is under vacuum, the door can't be opened), or if the system gets hotter than 200°C we don't want to accidentally oxidize the stainless with the inrush of oxygen.
- Min Vent Temp – the minimum allowable temperature in which the chamber can be vented. Reasons for this is usually condensation and ice. The inside of the chamber can frost/ice over if it is vented at sub-zero temperatures.

NOTICE

In the event that the cycle is completed or reseted and the chamber temperature is outside the Max/Min temperatures, the PLC will initiate a cycle to bring the temperatures to within venting range.

Characteristics for each zone:

A zone is defined by an object/area whose temperature is controlled by a thermal source such as a heater or LN2. Most systems only have one zone, but some have two or three. Three is the highest number for the standard software. An example of a single zone system would be a chamber with a shelf with a heater. An example of a two zone system would be a chamber with a platen and shroud AND they are controlled by two independent thermal sources. The below options are available for each zone the system has:

- SP Max: The maximum temperature setpoint for the zone.
- SP Min: The minimum temperature setpoint for the zone.
- Max Source: The maximum temperature of the source (heater or chiller). This acts as a limiting factor to protect product/equipment from getting too hot. When using a heater, if the heater temperature goes above the max source, then power to the heater will be cut off and a soft overtemp warning will be triggered.
- Min Source: Typically for Chillers. This is a limiting factor in preventing the chiller temperature from getting too cold.
- PID time: The PID sample time for the PID loops used for the zone.

PID Parameters for each thermal source.

Multiple thermal sources can be used in a single zone. Such as a heater/LN2 combination for a single platen. The heater and LN2 will run on separate PID loops. There are two selection boxes to choose which thermal source and which zone you want to define the PID parameters for.

Note: Laco chambers also have the ability to control the rate which the temperature can change over time. We call this the rate setpoint (or rate point). This rate point method

also runs on its own PID loop. Should the user decide to use this feature, the PID parameters need to be entered for the rate point.

NOTICE

Systems with chillers will not have PID parameters and these options will be hidden from view.

The PID Parameters:

- P-gain: This multiplies the error
- I-gain: This multiplies with the error summation.
- D-gain: This multiplies the PID derivative (rate of temperature change).
- The biases: The biases are used to predict the PWM at the steady state condition. For example, if you predict your chamber platen will hold steady at 80°C with a PWM of 40%, then the bias-PWM will be 40 and the bias-SP will be 80.
- Windup: This is used to determine if the temperature is in transition or at steady state. If the chamber is in transition, the integral will not be calculated. For example, if the user defines this value to be 0.3°C, then if the temperature changes more than 0.3°C per PID time, then the integral will reset to zero and not calculate.

To see how the PID loop is being calculated, see the maintenance section.

Sensor Labels:

Press this button to go to a screen to name the temperature sensors on your system.

Temperature Input Names

Input	Sensor Name	Value
Exp. Card1, 0	Frnt Rt	-58.6
Exp. Card1, 1	Bck Lft	-45.9
Exp. Card 1, 2	Bck Rt	-30.4

In this screen you can name each temperature being used. This name will show up on the run, manual, and calibration screens. It is limited to seven characters. Press the up and

down arrows to scroll through the different temperature inputs. Most systems will have three or less inputs.

4.3.4.3. User Configured Warnings and Faults:

	Enabled	Faultable	Time to Trigger	Time to Fault
Abort	☐	☐	00:00:00	00:00:00
Pump Down Duration	☒	☐	00:10:00	00:00:00
Vent 2 Atm Duration	☐	☐	00:00:00	00:00:00
Fill 2 Atm Duration	☒	☐	00:10:00	00:00:00
HV Duration	☒	☐	04:00:00	00:00:00
Soft Overtemp	☐	☐	00:00:00	00:00:00
Gauge Disconnected	☒	☐	00:00:10	00:00:00

This screen lets the user define what is a warning and what isn't. Here are some defined terms to understand how the warnings can be configured:

- **Enabled:** The warning is enabled on the system and the fault condition will not be ignored.
- **Active:** If a warning is enabled, the warning will be active when its criteria is met. For example, if the vacuum gauge is unplugged, then the "Gauge Disconnect" warning will be active.
- **Time to Trigger:** This is the amount of time a warning may be active before it will place the system into its warning state. When the system is in a warning state, everything will continue to function, but a warning message will appear on the screen.
- **Faultable:** This allows a warning to turn into a fault.
- **Time to Fault:** This is the amount of time it takes to fault the system after the warning has been triggered. When the system is faulted, all outputs are turned off and the cycle is ended. The fault condition must be remedied before you can clear it.

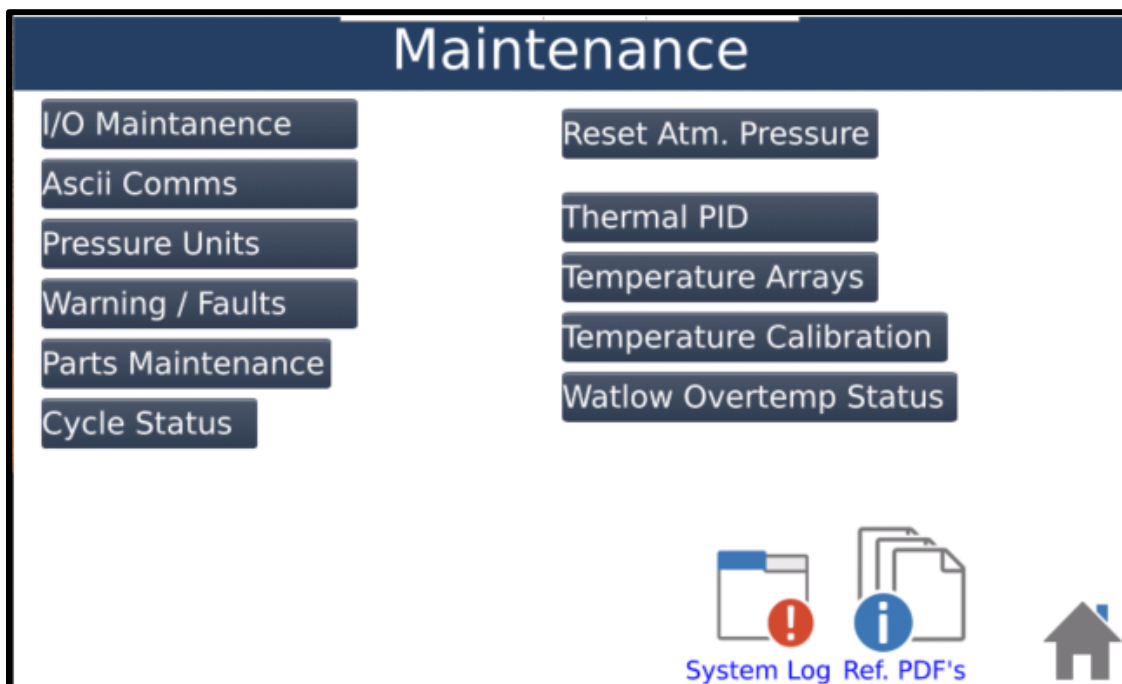
There are a number of user configured warnings on the system. These are optional warnings used to help the user control their process but do not endanger the system itself.

- **Abort:** Active when the abort button is pressed during a cycle.
- **Pump Down Duration:** For a non-high vacuum application, this is active while you're pumping your chamber down and have not reached your setpoint. For high

vacuum applications, it is the time it takes to open the gate valve. Example: the user knows it should only take 2 minutes to reach a 10 torr setpoint. They can enable this warning and set the time to trip to 3 minutes. But the user doesn't know that a newly installed fitting is leaking, and after pumping for 3 minutes the system will enter the warning state and notify the user.

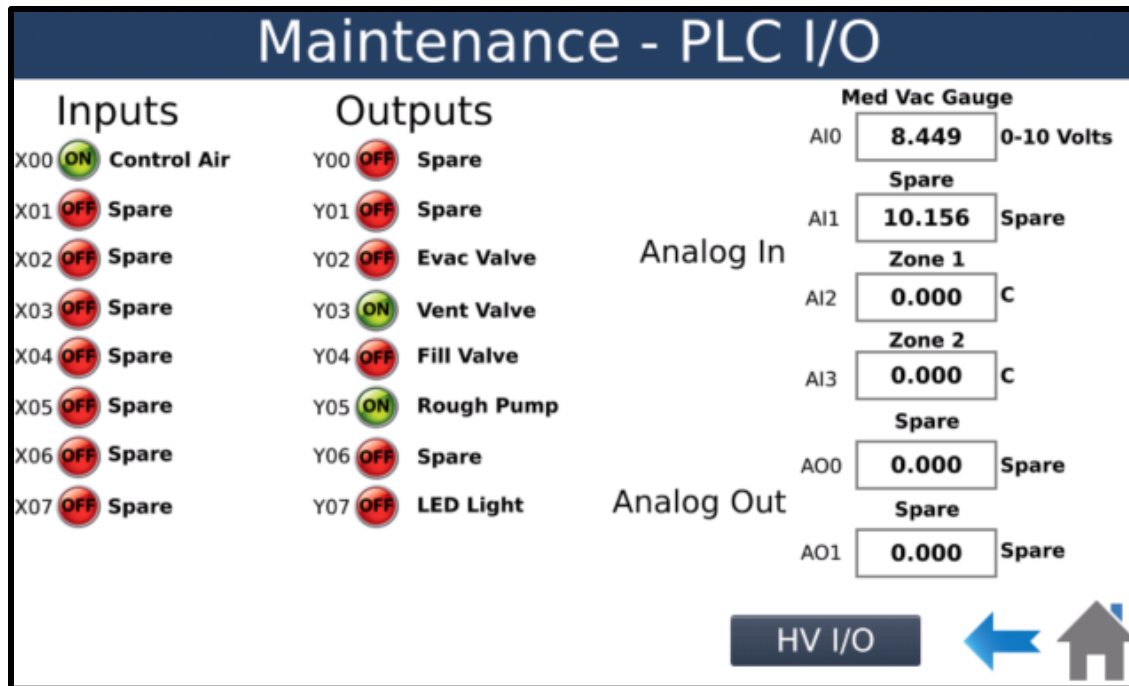
- Vent 2 Atm Duration: Active while venting to atmosphere
- Fill 2 Atm Duration: Active while filling to atmosphere (filling is defined as using a pressurized gas to back fill the chamber instead of room air). Example, the user is using a compressed N2 bottle to back fill the chamber. The user knows it should only take 2 minutes to back fill the chamber and so sets the time 2 trigger to 3 minutes. But the user doesn't know the N2 bottle is empty and it doesn't fill the chamber. After 3 minutes, the system will enter a warning state and notify the user.
- HV Duration: Active when the gate valve is open and the pressure is above the setpoint. Example: The user knows the system should be able to reach 7E-6torr in 2 hours, so 2.5 hours is entered into the time to trigger field. If the system can't reach its setpoint in 3 hours, the test must be aborted, so 30 minutes is entered into the "time to fault" field and the "faultable" box is checked. The user places a product in the chamber that wasn't cleaned properly and has a heavy gas load. After pumping for 2.5 hours the system goes into the warning state and notifies the user. After another 30 minutes the system faults and ends the cycle.
- Soft Overtemp: Active when the heater temperature is greater than the maximum source temperature (see section 6.3.4.2).
- Gauge Disconnected: Active when a gauge is unplugged.

4.3.5. Maintenance Screens



The maintenance screens are there to help troubleshoot issues that may arise. The menu options are configuration dependent. The options shown above does not represent all options, and your system may not have all the options shown above.

4.3.5.1. IO Maintenance



The I/O maintenance screens show you the status of the digital I/O and the analog inputs. The user can use these screens to help trouble shoot problems.

4.3.5.2. ASCII Communications Maintenance

ASCII Comms

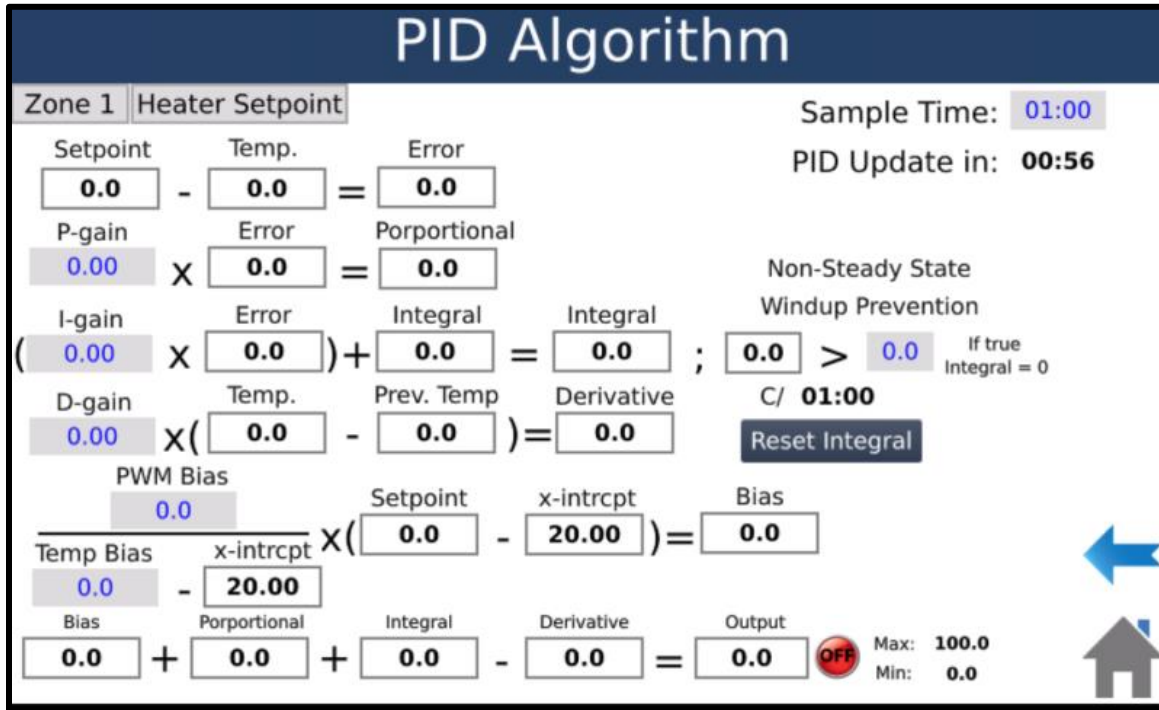
RS-232 Data In	RS-232 Data Out
?VP{CR}	VP:6.51E2{CR}
TCP Data In	TCP Data Out



Ascii Code Instructions

ASCII codes are used when communicating via RS232 and Ethernet. The above screen will display the last code sent to the PLC, and also show the response back to your computer. The Ascii Code Instruction Icon will open a PDF document that shows you the available commands to use.

4.3.5.3. Thermal PID Algorithm



The PID maintenance screen shows you the math being used by the PLC to controller the temperature setpoint. The seven highlighted boxes are the tuning parameters defined in the advanced settings and are configurable in this screen. The user must be logged in with the advanced password to change these values.

You can toggle through the zones by pressing the top left button (says zone 1 on screen above). And the user can toggle through the PID algorithms for each zone, which depending on the system can be any of the following:

- Heater Setpoint
- Heater Rate
- LN2 Setpoint
- LN2 Rate
- Chiller Setpoint
- Chiller Rate

4.3.5.4. System Pressures, shows the chamber pressure in all the units.

Pressures

Torr (mmHG)	652.2166748	meters	1271.595
PSIA	12.6138697	feet	4171.886
PSIG	0.0000000	miles	0.790
inHG_A	25.6777706		
inHg	0.0000000		
mbar	869.5483398		
Pascal	86953.5312500		
atm	0.8609260		
hPa	869.5483398		
kPa	86.9548340		



4.3.5.5. System Warning/Fault Status

User Warnings/Faults

	Enabled	Active	Active Time	Time to Trip	Tripped	Faultable	Fault Time	Time to Fault
Cycle Abort	OFF	OFF	00:00:00	00:00:00	OFF	OFF	00:00:00	00:00:00
Vacuum Setpoint	ON	OFF	00:00:00	00:10:00	OFF	OFF	00:00:00	00:00:00
Vent To Atm.	OFF	OFF	00:00:00	00:00:00	OFF	OFF	00:00:00	00:00:00
Fill To Atm.	ON	OFF	00:00:00	00:10:00	OFF	OFF	00:00:00	00:00:00
Gauge Disc.	ON	OFF	00:00:00	00:00:10	OFF	OFF	00:00:00	00:00:00
High Vac Setpoint	ON	OFF	00:00:00	04:00:00	OFF	OFF	00:00:00	00:00:00
Soft Overtemp	OFF	OFF	00:00:00	00:00:00	OFF	OFF	00:00:00	00:00:00

System Warning/Faults


This screen will display the status of the system warning/faults. It will show which ones are enabled, which ones are active and which ones are tripped. Press the “System Warning/Faults” to view the status of the non-user configured warnings/faults.

4.3.5.6. Recipe Cycle Status

Recipe Cycle Status

Pressure Setpoint:	20.00	Torr	Hold Time:	Hold Timer:
Current Pressure:	666.6	Torr	00:00:00	00:00:00
Lower Limit:	0.00	Torr		
Upper Limit:	0.00	Torr		

Pressure Status: Pressure Not In Range

Skip to Next Step

Zone 1

Temperature Setpoint:	0.0
Current Temperature:	0.0
Lower Limit:	-3.0
Upper Limit:	3.0
Temperature Status:	N/A



Use this screen to check the status of your recipe cycle. If running a cycle seems to get stuck on a specific step, check this screen to see why. It will tell you if the vacuum and/or temperature is out of range. If for whatever reason the user decides to move on to the next step, the user can press the “Skip to Next Step” button to move on.

4.3.5.7. Temperature Input Calibration

Temperature Calibration

Reset Calibration

Select the Sensor to Calibrate

Sensor Name	Input	Value
Bck Lft	Exp. Card1, 1	-45.7
Bck Rt	Exp. Card 1, 2	-30.2
Frnt Lt	Exp. Card1, 3	-65.0

Calibration Slope: 1.0000

Calibration Intercept: 0.000

$\text{Slope} \times \text{PLC Reading} + \text{Intercept} = \text{Calibrated Temperature}$

1.0000 X -30.2 + 0.000 = -30.2

The user can calibrate their temperature sensors if they want to. Select which sensor you want to calibrate using the scroll arrows. The sensor in the middle of the scroll box is the sensor being calibrated. In the “Calibration Slope” and “Calibration Intercept” boxes you can enter your calibrated slope and intercept parameters.

The equation on the right will show you what the PLC is reading, and what the new calibrated value is. Pressing the reset button in the top left will reset all temperature slopes to 1, and intercepts to 0 (this means the calibrated temperature is equal to what the PLC is reading). This button is only available if you are logged into the advanced settings.

How to Calibrate

You’ll need some kind of known reference temperature. Take the RTD sensor and place it in the known temperature source. Record the PLC reading and the actual temperature. Repeat this process for other temperatures across your operating range (at least two temperatures, but more is better).

Run a linear regression on your data using excel or other preferred means to generate a slope and intercept. Enter the slope and intercept values into the grey boxes on the above screen.

4.3.5.8. System Logs

System Log		
SystemHistory		Enter Array On
#	Time	Message
0	01/17/19, 09:41:25 PM	Cycle Complete
1	01/17/19, 09:32:35 PM	Start Recipe: ASTM6653
2	01/17/19, 09:27:02 PM	Gauge Reconnected
3	01/17/19, 09:26:58 PM	Fault: Gauge Disconnected
4	01/17/19, 09:26:14 PM	Warning - Low Air Pressure
5	01/17/19, 09:24:51 PM	Start Recipe: feet

The system logs page displays a time stamped log of the last 40 faults/warnings and cycles started and completed on this system.

4.3.6. Data Log Screen

Data Log Settings

USB drive present. Data will be auto transferred to drive.

Data Log Interval: 00:20:00 Data Log Method: Manual Data Log

Select desired data table and action button below

Recipe Steps

Export Data Table to SD Card

Import Data Table from SD Card

Convert CSV file to Data Table

SD Browser

Home icon

All 3500 controllers will ship with an internal micro SD card. Users can log data to the SD card in CSV file format by selection one of the following 3 data log methods.

- None
- Manual Data Log – When this option is selected the user will see a disk icon on both run and manual screens. When the disk icon has a red check mark no data is being logged. The user must select the icon and data will start to log and icon will have a green check mark. Press the button again to stop logging data.
- Process Data Log – For this option data will automatically be collected during automated process cycles.

Select the desired data log interval (in HH:MM:SS format) for data to be logged. If a user inserts USB disk drive (DOK) into the USB port and whenever data logging is finished (end of process or press the disk icon) the data log CSV file will be automatically transferred from the SD card to the USB drive. After data logging is finished recording, the user should wait several seconds before unplugging the USB drive. If you unplug it too early the data won't transfer. If this happens the data will still be retained on the micro SD card.

If the a USB drive is not attached the user will see a yellow banner per the screenshot below. Data logging will still be stored on SD card. To retrieve the CSV file user must attach a USB and go into UniApps ([see section 9.2](#)). The user will go to the memory tab and file browser and click on the data tables folder on the SD card. Once the desired files are selected the user can manually transfer the files to the USB drive by selecting the transfer button. To remove the USB drive from the controller the user must click the

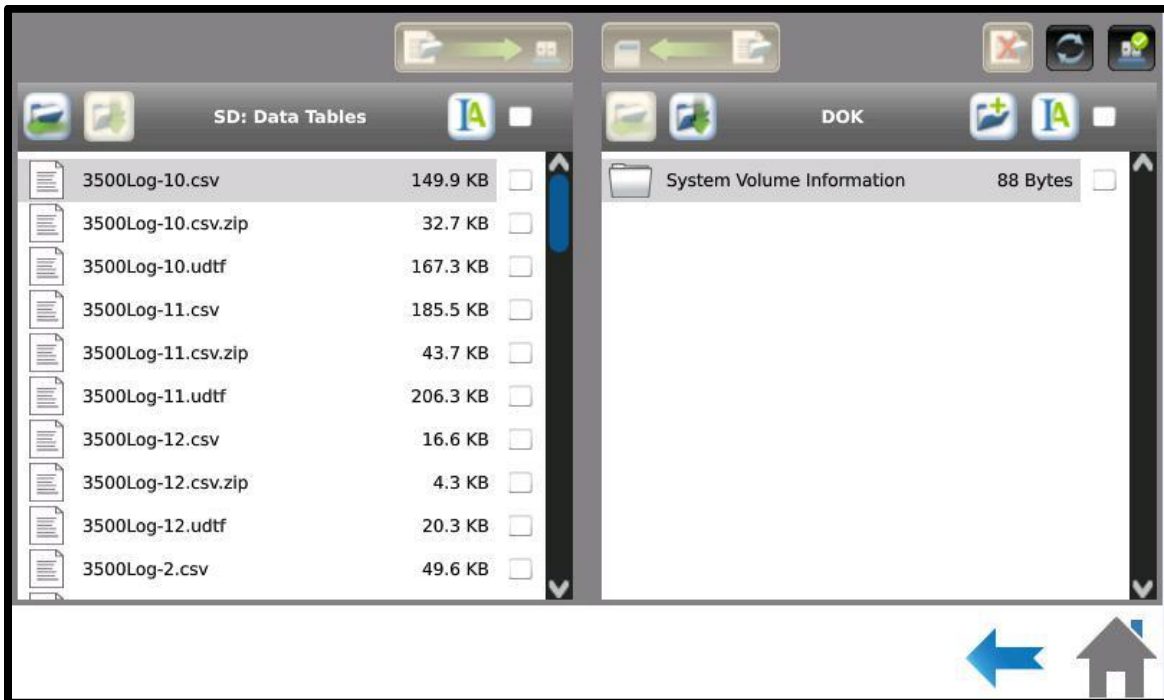
eject USB button. It is recommended to delete old files off the SD card periodically as user has retrieved and saved the files offline.

CSV files are saved in the following format: “3500Log-#.csv. The # will be a number that will increment after each data logging run. For 3500 systems with multiple chambers the user file format will be “3500Log2-#.csv for chamber 2 and so forth.

This screen also gives the user to export/import data tables to/from the SD Card. The data tables contains various information used to store information about the system and helps the system run properly. The following tables are available to save on the SD Card:

- Recipe Steps – Steps in each recipe
- Advanced Settings – The configuration of your advanced settings
- System Logs – List of the previous 40 Status Messages
- Recipe Info – Recipe names

If you want to view the files on the SD card and the jump drive (if inserted), then press the “SD Browser” button:



In this file browser window, you can view the files on the SD card and transfer them to the jump drive.

Example: transfer system logs to a CSV file on a thumb drive:

1. Select System Logs from drop down menu
2. Press the “Export Data Table to SD Card” button.
3. Press the “SD Browser” Button
4. On the left window screen, select the “Data Tables” folder and press the open folder button (top left side of screen)



5. Select the file “3500SystemLogs” and press the transfer to DOK button

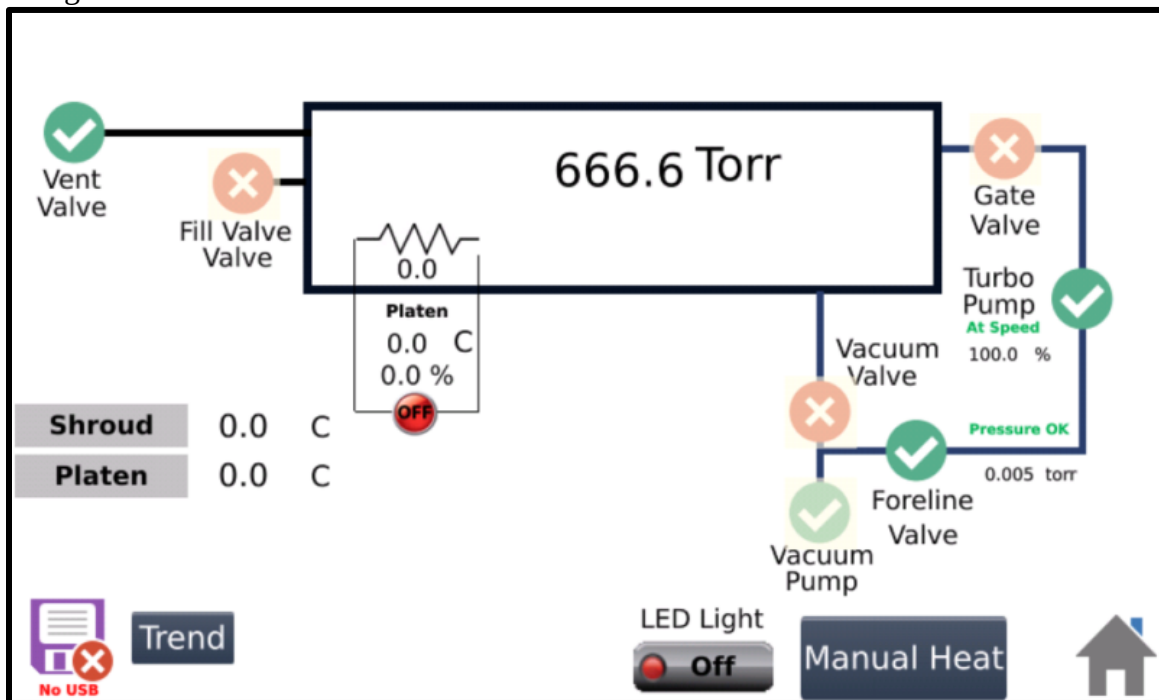


6. Typically, you should wait about 1 to 2 minutes for the transfer to be completed

4.3.7. Manual Mode

Manual mode is a powerful tool to troubleshoot, test and understand your 3500 system. To enter a manual mode a technician or advanced level user must press the manual icon from the home menu.

Touch the selected system icon to enable valves, pumps or outputs. When any output is off, there is a red X on it and when any output is on, there is a green check on it. All manual mode screen options have interlocks built into the system. If all conditions for the device have not been met, the icon will be greyed out and will not change. If you press the icon anyways, a message will appear in the top left corner explaining why you can change the state.



A valve icon will be greyed out if an interlock is preventing the valve from opening. The interlocks are as follows: (note that some items are only applicable to high vacuum systems).

- Evac: Cannot be opened if any of the following conditions are not met:
 - Vacuum pump must be on for 10 seconds
 - Vent, Fill, Foreline and Gate valves must all be closed
- Vent: Gate and Vacuum valves must be closed
- Fill: Gate and Vacuum valves must be closed

- Foreline Valve
 - Vacuum Pump must be on for 10 seconds
 - Vacuum Valve must be closed
- Turbo Pump:
 - Foreline Valve must be open (it does not need to stay open after the pump is on)
 - Foreline Pressure must be in the “OK” region. The foreline is “OK” once it reaches 0.2 torr, and stays “OK” until it gets higher than 0.5 torr
- Gate Valve:
 - Chamber pressure must be below the gate cross over pressure (usually 0.5 torr).
 - Turbo Pump must be on.
 - Foreline Valve must be open.
- Vacuum Pump: Can turn on at any time. Cannot turn off if foreline valve or evac valve is open.

If the system is a thermal system, either through heaters, chillers or LN2, to set the temperature setpoint, press the “manual heat” button. See below for instructions on that screen. Once you have entered a temperature setpoint and activated the heaters, you’ll see in the above screen the temperature, setpoint, and the operation of the thermal source.

4.3.7.1. Manual Mode – Thermal Manual Screen

Manual Heat		
Platen Temp Setpoint	20.0 °C	Thermal Control On
Shroud Temp Setpoint	20.0 °C	Thermal Control Off
Aux. Platen Temp Setpoint	20.0 °C	Thermal Control Off
Platen Temp Rate	0.0 $\frac{^{\circ}\text{C}}{\text{min}}$	Rate Control Off
Shroud Temp Rate	1.0 $\frac{^{\circ}\text{C}}{\text{min}}$	Rate Control Off
Aux. Platen Temp Rate	0.0 $\frac{^{\circ}\text{C}}{\text{min}}$	Rate Control Off



The Manual Thermal screen allows the user to manually set temperature setpoints and run to those values. A user can enter a setpoint and press the temp control button to on and the system will automatically ramp to the setpoint value.

The user can also control the rate at which the temperature can change. The rate value must be set, and the rate control button must be pressed on before turning on the thermal control. If the thermal control is on, the ability to turn on rate control is disabled.

Note: Controlling the thermal rate of change can be difficult. It works best for heaters. It can struggle while cooling with chillers.

4.3.7.2. Recipe Screen

Recipe - Vacuum & Thermal						
Recipe: 1 - Water Mark				Last Step Number: 3		
Step #	Vacuum Step	Vac SetPoint Torr	Hold Time	Hold Step	Optional Step	Zone1
0	Setpoint	7.00E-6	00:00:00	☐	Off	0.0
1	Setpoint	7.00E-6	00:00:00	☐	On	80.0
2	Setpoint	7.00E-6	00:00:00	☐	On	-50.0
3	Cycle	Step 1 X 5	00:00:00	☐	Off	0.0
4	Setpoint	0.00	00:00:00	☐	Off	0.0



Thermal Settings 

Press the recipe icon on the home screen while in technician or advanced mode to access the recipe screen. The 3500 controller allows up to 20 recipes with a max of 50 steps each. Each recipe can be given a name to identify the process to be run.

The user can program the steps, setpoints and timers for their desired process. Only the top row of the recipe steps can be edited while the controller is not running a process. The lower four rows are shown for convenience so one can see subsequent recipe steps. Press the next or previous arrow keys to scroll through and edit the needed recipe steps.

Vacuum Step

Press the vacuum step screen to toggle between the states listed below. All the states listed below may not be available depending on the users configuration.

- None – This step closes all valves for the entire step.
- Setpoint – Use this feature when you want to pump/fill to a specific setpoint.

- Vent2Atm – This opens the vent valve until atmospheric pressure is reached. The vent valve will remain open until the next step
- Fill2Atm – This will open the pressure valve (if included). The pressure valve will remain open until atmospheric pressure is reached, and then it will close.
- ASCII – Use this step when the user wants to control the process from a computer using RS232 communications with ASCII commands. (See appendix for list of commands)
- ATM Purge – This will fill the chamber with N2 gas until it reaches atmospheric pressure. Once it reaches atm, the vent valve will open. N2 gas will continue to go into the system and out the purge valve.
- Cycle – This allows you to return to a previous step and repeat the steps in between. For example, in the screen shot above, step 3 is a cycle step. It shows “Step 1 x 5”. This means that once the program reaches step 3, it will return to step 1 and then proceed on to 2 and then 3 again. It will repeat this process 5 times.

Vac Setpoint

Press the vacuum setpoint and the “Recipe – Vacuum Setpoints” screen will show up. See below for these screens.

Hold Timer

Enter the desired hold time to run once the setpoint is achieved. If a hold time of zero is entered, then as soon as the setpoint is reached the process will proceed to the next step. When a hold timer is running and the process goes outside the control band then the hold time pauses and keeps the accumulated time. Once the process is back in the defined control band the hold time resume where it left off.

Hold Step

This indicates a holding step. Once all setpoint requirements are met, and the hold time has reached zero, the system will continue to hold this state until the user presses the start button

Last Step Number

This indicates which step the user wants to be the last step. In the above screen, the last step is 2. This means that when step 2 is completed, the cycle will end.

4.3.7.3. Recipe – Vacuum Setpoints

Recipe - Vacuum Setpoints		
Parameter Options	Setpoints	Units
Pressure Setpoint	7.00 E 1	Torr
Control Range	2.000	Torr
Use Vent Valve to Repressurize ☐ (Pressure valve will be used if left unchecked)		



In this screen you define your vacuum setpoint. If it is a high vac system, the setpoint will be displayed as a mantissa/exponent format. If its not a high vac system, it will be displayed as a regular decimal number.

4.3.8. Run Screen and Running processes.

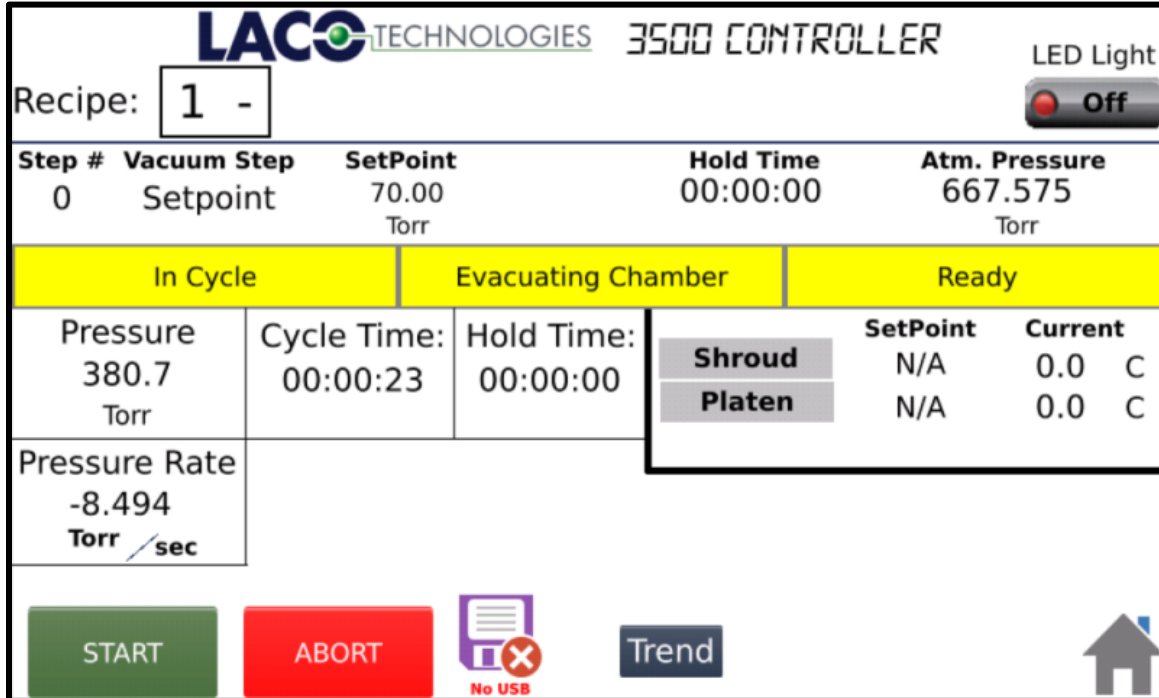
When the 3500 controller is powered on the information screen is shown for 4 seconds and then the run screen appears. The run screen allows the user to run automated processes. The run screen may vary depending on the user configuration. Shown below are the run screens for non-thermal and thermal applications. If a custom run screen is used to match the customer's application refer to the 3500 system manual.

Once a correct recipe has been entered and verified the user can start a process by pressing the start button. Ensure the vacuum system is ready with the door closed when a process is started. While the process is running the current step recipe information will be shown in the top of the screen. Below this section, colored background a status message bar is displayed with the following possible messages.

- Process Recovery
- Process in Cycle
- Process Complete / Ready For Start
- Press Start to Continue – This message displays on “hold” steps when all setpoints are satisfied. The user must press the start button to continue the process.

Listed below the status bar are various live pressure and temperature readings and timers. The information display will vary depending on the user configuration.

To abort or stop the process while currently running press the abort /stop button. A message box will appear to confirm that the user wants to abort the process. Once the process is aborted the recipe advanced to the start home step to take the system back to ready state.

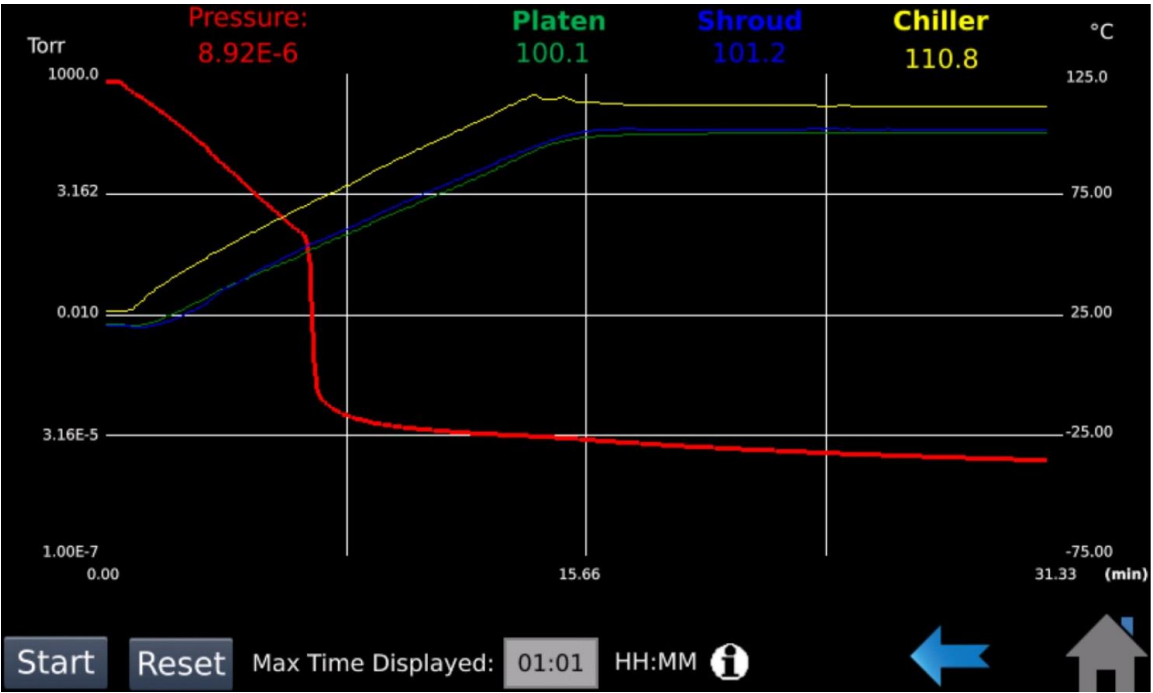


For thermally controlled systems, up to three temperatures are available for viewing. You can select which temperatures are viewed by pressing the grey sensor name box to the left of the temperature readings. This will switch the sensor you would like displayed on the screen. Continue pressing the name until the sensor you want to see shows up.

4.3.9. Trending Graphs

You can get to the trending graph either by the run screen or manual mode screen. Press the trend button to access it. The temperatures that show up on the trending graph are the three (or less) temperatures that show up on toggle options on the run or manual screens.

- Start: begins to record pressure and temperature
- Reset: stops recording and erases the data
- Max time displayed: this is the maximum time duration that will be displayed on the graph. Once the recording time reaches this time limit, the x-axis starts to move so only that time duration is shown.
- Go to advanced settings to adjust the y axes. Note that adjusting the y-axes does not auto scale the graph. So the reset button should be pressed before re-scaling the axes.



5. Remote Connection

The VC/HVC3500 controller has the ability to be controlled remotely and have the files in the SD Card accessed via computer.

5.1.1. Connect your computer to the VC/HVC3500 Controller

First you will need to connect your computer to the VC/HVC3500.

1. Obtain the Panel IP Address from the VC/HVC3500 controller.
 - a. See section 6.4 on how to obtain this.
 - b. If the user desires to change the IP address, please check with Laco Technologies first, as there may be other items on the system that will be effected by changing the IP address.
2. Configure the ethernet IP settings on your computer so you can connect to the IP address listed on the Panel IP settings. You may need to contact your I.T. department for assistance with this.
3. If you just want your VC/HVC3500 to be local for a specific computer, then plug the ethernet connection from the computer to the ethernet port on the side of the electrical enclosure (RJ45, labeled Ethernet).
4. If you want to place the VC/HVC3500 controller on your company's network, connect the "Ethernet" port to your company's internet. Get with your I.T. department to properly route the VC/HVC3500 controller to your network.

5.1.2. Controlling the VC/HVC3500 remotely

The user can control the VC/HVC3500 controller via their computer, which enables monitoring and control at a cubical or even at home. Follow the below steps to set up your computer to do this:

1. Ensure your computer is connected to the VC/HVC3500 controller
 - a. Easiest way to do this is to ping the IP address.
 - b. See the previous section on how to connect to the controller.
2. Install any VNC Viewer software onto your computer.
 - a. Laco uses Real VNC Viewer for internal testing and applications. At the time of this writing, Real VNC Viewer is the first google result when searching "VNC Viewer".
3. Open the VNC Viewer, depending on which VNC Viewer software you have, you should just need to enter the Panel IP address of the controller, and you will be connected.
 - a. When connected, you'll see the HMI screen, and you'll be able to click on buttons as if you were standing at the machine.

5.1.3. Remote accessing files on the SD card in the VC/HVC3500 controller.

The user can access the files on the SD Card via their computer. This is way more convenient than transferring files to a flash drive or pulling the SD Card out. The files are accessed via an FTP Server.

1. Ensure your computer is connected to the VC/HVC3500 controller
 - a. Easiest way to do this is to ping the IP address.

- b. See section 5.1.1 on how to connect to the controller.
2. Download and install any FTP Client software onto your computer. Laco Technologies uses FileZilla for internal testing and applications. At the time of this writing, FileZilla is the first google result when searching “FTP Client”.
3. Once the FTP Client software is installed and opened, use the following settings:
 - a. Host: Panel IP Address
 - b. Username: Laco5226 (case sensitive)
 - c. Password: Laco5226 (case sensitive)
 - d. Port: 21
4. You should now have access to the files on the SD Card. The data log files are stored in the “DT” folder. Screen shots are saved in the “Media” folder.

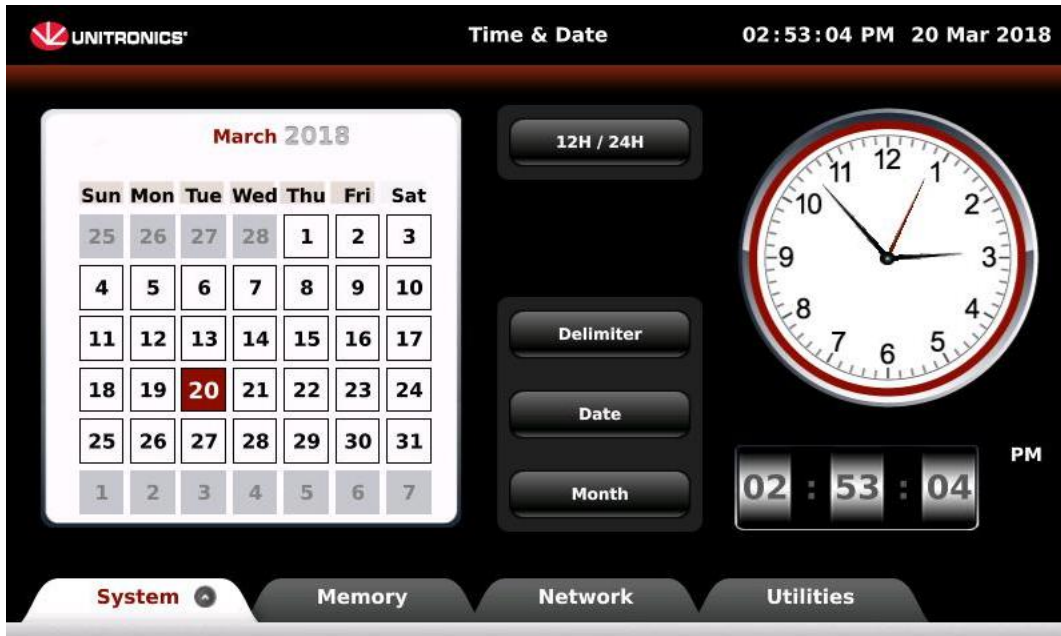
6. Advanced Unitronics Settings and Downloading System Updates

By pressing in the upper right corner of any screen for more than 2 seconds, system users can access advanced Unitronics controller features. A user will then select the UniApps icon and the UniApps main menu screen will appear. When the user is done using UniApps the user will press the upper right area again and then return to the user application.



6.1. Set the Date and Time

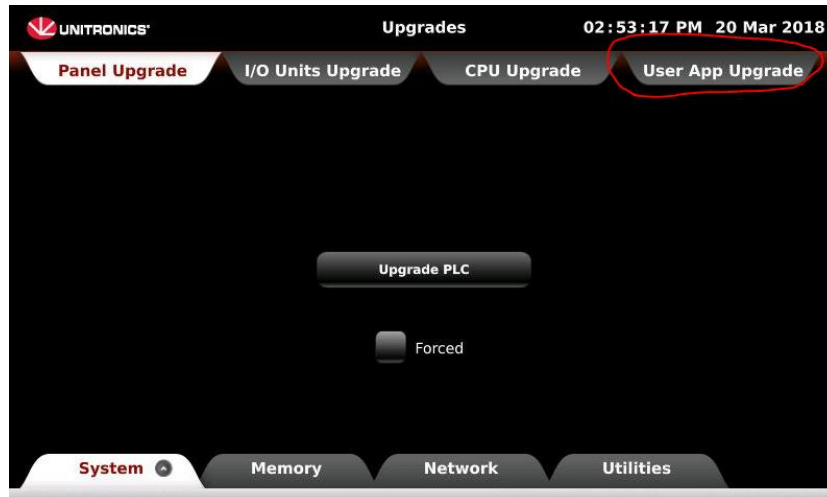
To change the date and time of your controller, press the “Time & Date” icon on the advanced unitronics home screen. This will take you to the window that will let you set the day and time.



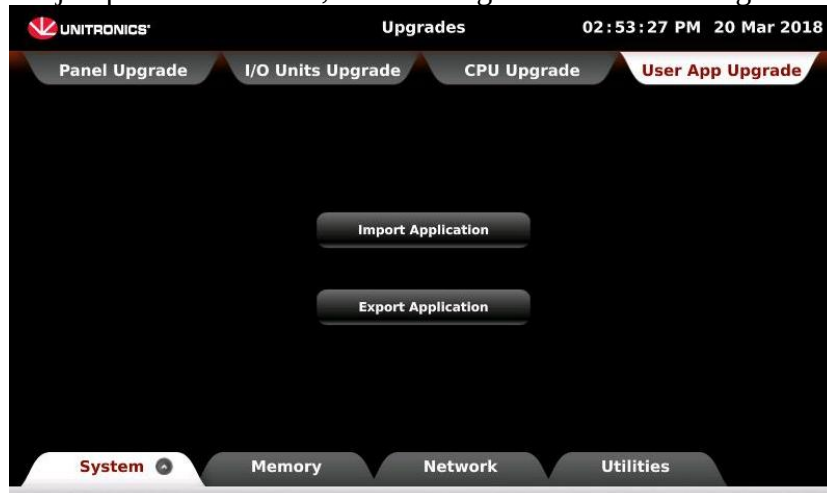
6.2. Software Update

Sometimes Laco will give a software update to a customer for various reasons which include customers requesting for additional features in the software. See below for the instructions on how to download the software update to your PLC.

1. Make sure jump drive is formatted to FAT32
2. An email with a .zip file with the update will be sent to the customer.
3. Unzip the file and place the resulting folder on a jump drive.
 - Note: sometimes unzipping the file will rearrange the file path. Make sure the file path on the jump drive is as follows:
 - Removable Disk/UniStream_Files/UserApp/userApps_USP-070.zip
 - If the path on your jump drive is different, rearrange so its located properly.
4. Plug the jump drive into the usb port on the side of the controller.
5. On the Uni Apps Home Screen, select the upgrade icon.
6. Select the “User App upgrade” on the top right corner of the Upgrades screen.
(Do not press “Upgrade PLC”)



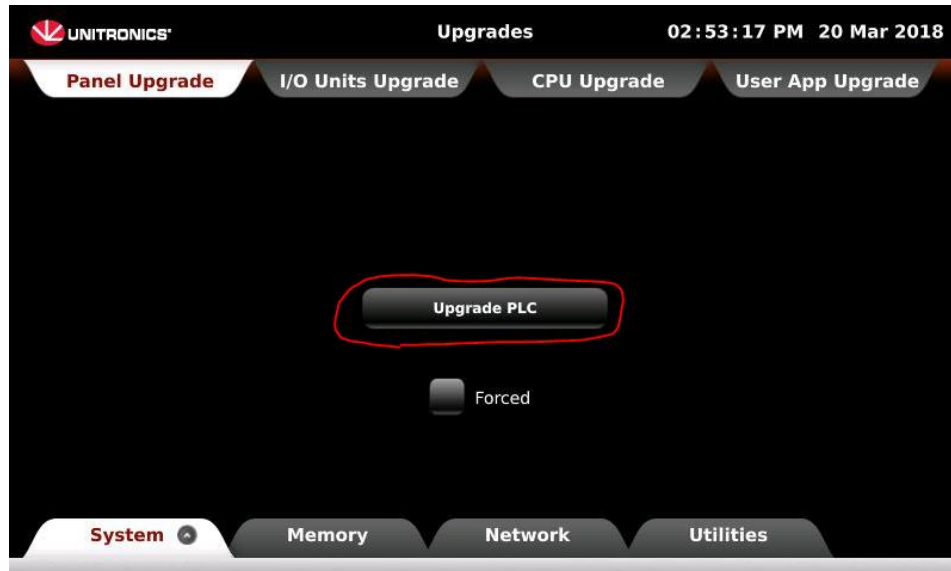
7. With the jump drive plugged in, press “Import Applications”. If the file path in the jump drive is correct, it will recognize the file and begin the download.



6.3. Firmware Update

Sometimes a software upgrade will also accompany a firmware update. Laco will send a .zip file through email with the updated firmware.

1. Make sure jump drive is formatted to FAT32
2. Unzip file and place on jump drive.
3. The folder path should be: Removable Disk/Unistream_Files
4. Place the jump drive into the USB port on the side of the control box.
5. On the Uni Apps Home screen, select the “Upgrades” icon.
6. On the “Upgrades” window, select “Upgrade PLC”



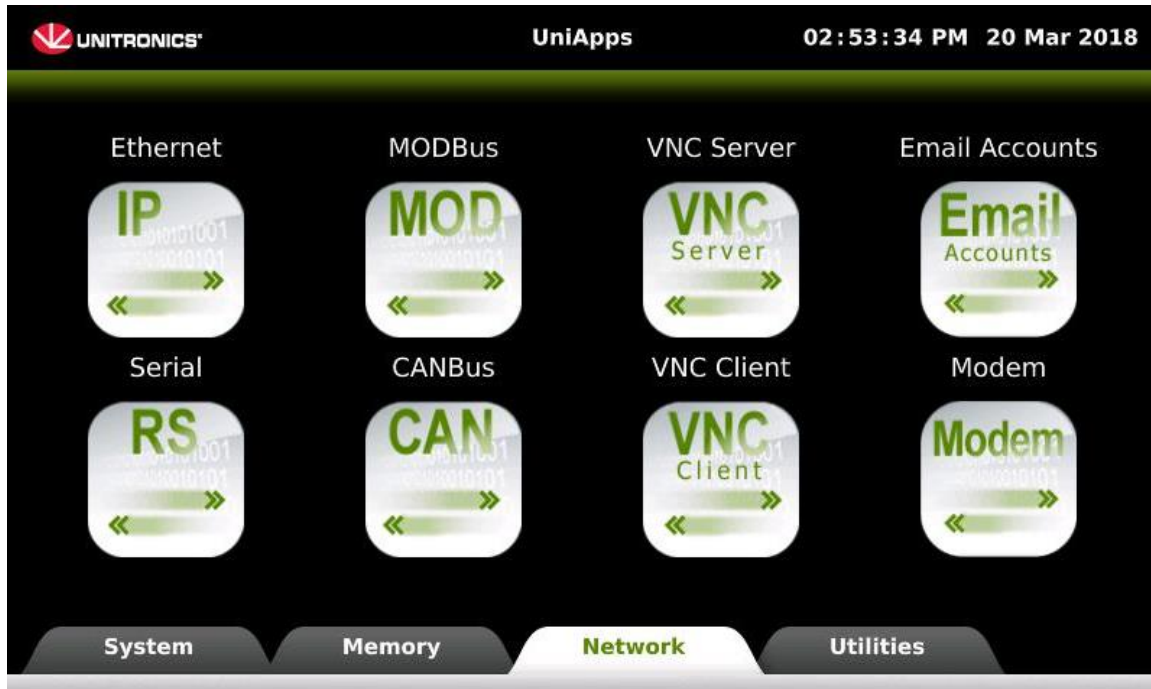
7. If the file path is correct, the firmware will begin downloading. This will take about 15 minutes.

6.4. IP Addresses

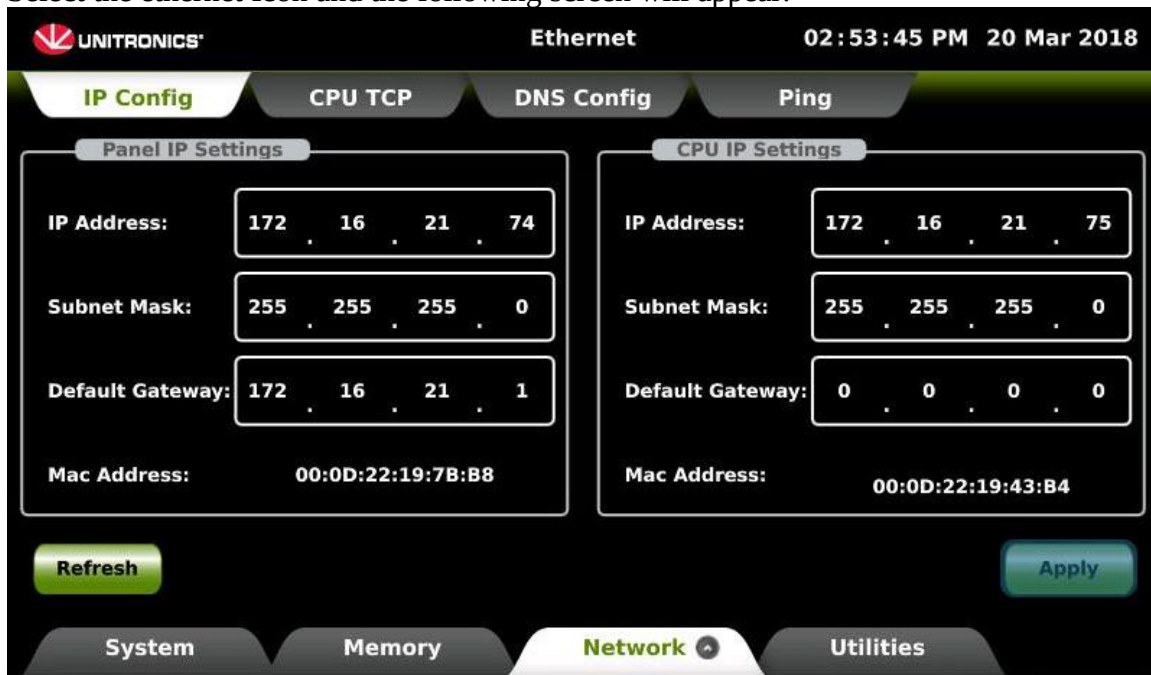
The Unitronics controller has two IP addresses: one for the Panel, the other for the CPU. The IP address for the panel is used to connect to the actual program loaded on the PLC. The CPU address is used for communications via Ethernet/TCP. Typically the end user will only need to change the CPU address to communicate with devices at their facility. The RJ45 jack on the side of the control box can connect to both the panel, and the CPU.

6.4.1. Changing the IP Address

On the Uni Apps home screen, select the Network tab in the bottom middle of the screen and the network home screen will appear:



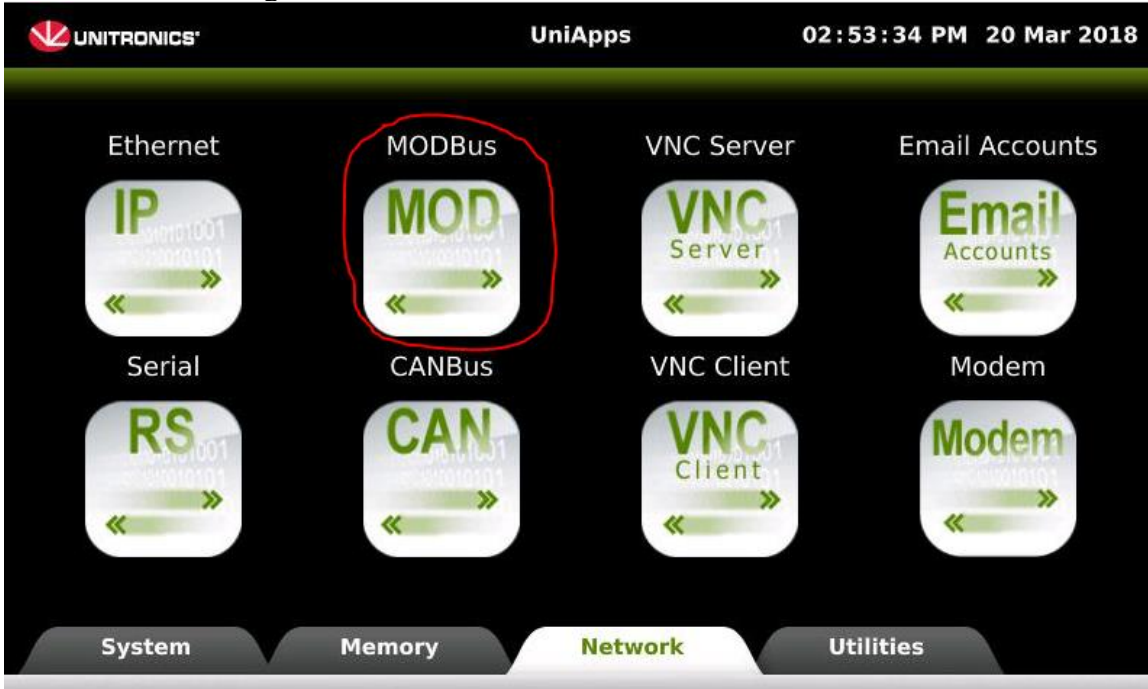
Select the ethernet Icon and the following screen will appear:



Enter the new IP Addresses and press the Apply button when your finish. Remember, use the CPU IP address to communicate with a computer via Ethernet/TCP using Ascii code. Note, changing the IP address may have adverse effects on the system. Please contact Laco Technologies before changing your IP Address to make sure you don't lose communication with remote I/O or other devices.

6.4.2. Change IP Address for Third Party Devices

On standard systems a third party device is typically not used. However, sometimes the end user request such devices be placed on the network and communicate to it via various protocols such as Modbus. The IP address for these devices are assigned at the device itself, but for the PLC to communicate with it, the IP address needs to be made known to the PLC. To do this go to the Network home screen and select the “Modbus” Icon:



This will bring you to the screen that list all Modbus devices on the ethernet network, along with its slave ID, IP address , etc. You can enter the IP address of the third party device here.

UNITRONICS

MODBus

02:54:21 PM 20 Mar 2018

Ethernet Master

Ethernet Slave

Serial Master

Serial Slave

Slave ID	Alias	IP	Port	Status	Sessions	Successes	Failures	Dropped
6	Honeywell	172.16.21.76	502	1	46716	46716	0	0

System

Memory

Network

Utilities

7. Faults, Warnings and Troubleshooting

When a fault occurs the control system stops all processes and the fault description will be displayed on the test screen status with a red background. To recover from a fault press the red Abort/Reset button. The system will then proceed through the “home steps” back to a ready to test state. When in a fault state the user can go to manual mode if desired.

When a warning occur it means something doesn't seem right. The process will continue, but the user should investigate to make sure everything is working properly. All timers can be adjusted in the advanced settings menu.

Refer to the table below for help in identifying fault/warning causes with likely remedies to fix fault causes. There are both global or system faults and process related faults. Global faults (the first 2 faults in the table) can occur at any time whereas process faults only occur at certain steps in the test process that are not function specified. Note that fault histories can be reviewed on the fault history page.

No	Fault Description	Fault Causes	Fault Troubleshooting
0	Abort	The Abort Button was Pressed	• User Configurable Fault • This fault can be configured to be disabled.
1	Evac/Rough	The system is unable to evacuate to the vacuum setpoint (of gate Xover pressure) before the evac delay timer expires.	• User Configurable Fault • Ensure there are no vacuum leaks on the chamber or system. • Ensure the vacuum pump is performing as required and is properly maintained. • Evac fault timer may need to be increased for long evacuation times. • This fault can be configured to be disabled.
2	Vent Valve / Vent 2 Atm	The system is unable to vent to the vacuum setpoint before the vent warning timer expires.	• User Configurable Fault • Ensure the vent valve inlet is not blocked or plugged. • Vent fault timer may need to be increased for long vent times. • This fault can be configured to be disabled.
3	Fill Valve / Fill 2 Atm	The system is unable to pressurize to the vacuum	• User Configurable Fault

No	Fault Description	Fault Causes	Fault Troubleshooting
		setpoint before the pressure warning timer expires.	• Ensure the pressure source regulator is set to slightly higher pressure than the pressure setpoint needed. • Ensure Pressure source (such as compressed N2 tank) is not empty • Pressure fault timer may need to be increased for long pressurization times. • This fault can be configured to be disabled.
4	High Vacuum Warning	The system is unable to reach the high vacuum setpoint within the prescribed amount of time.	• User Configurable Fault • Ensure there are no vacuum leaks on the chamber or system. • Ensure the chamber is clean and dry • High vacuum time may need to be increased. • This fault can be configured to be disabled.
5	Soft Overtemp	The heater(s) are hotter than the “Max Source” setpoint in the advanced settings.	• User Configurable Fault • Check to make sure your setpoint is not higher than the max source point. • Review PID parameters, if needed, contact Laco for help. • This fault can be configured to be disabled.
6	Gauge Disconnected	One of the system gauges has been disconnected	• User Configurable Fault • Check the gauge cables and verify they are plugged in. • This fault can be configured to be disabled.
16	Control Air	The control air pressure dropped below an acceptable level. Valves may not operate	• Make sure shop air is connected. • Increase air pressure on main control air regulator to at least 15 psig above the pressure switch setpoint.

No	Fault Description	Fault Causes	Fault Troubleshooting
17	Hard Overtemp	The heater temperature exceeded the setpoint on the watlow overtemp controller. Will not automatically reset	- Make sure the setpoint and the “max source” setpoint are not above the max setpoint on the watlow controller. - Need to physically reset the watlow controller to clear error.
18	Gate Valve Not Opening	The gate valve is not opening when it should.	- Make sure there is air pressure on the system. - Verify the limit sensors on the gate valve. The valve may be open, but the sensor may have moved to the wrong spot.
19	Gate Valve Not Closing	The gate valve is not closing when it should	- Make sure there is air pressure on the system. - Verify the limit sensors on the gate valve. The valve may be open, but the sensor may have moved to the wrong spot.
20	Chiller Comms	The HVC3500 controller has lost communications with the chiller	- Check the communication cable between the HVC3500 and the chiller. - Verify communication settings (IP Address, Serial baud rates, etc). Contact Laco for support with this.
21	PLC Comms	Communication between the HMI and the I/O PLC (renu) have been lost	Contact Laco for support
22	HV Box Comms	Communication between the HVC3500 and the high vacuum enclosure have been lost	- Check the communication cable between the two enclosures. - Contact Laco for support
23	Turbo Fault	The turbo is in an error state	Check the manual for the turbo pump

No	Fault Description	Fault Causes	Fault Troubleshooting
24	Door Open	The chamber is trying to pump down but there is a massive leak (likely being caused by the door being open).	• Verify the chamber door is closed. • Check all fittings on the chamber, verify there all fittings are present and there are not open holes.
25	Foreline Pressure	Warning occurs when the pressure in the foreline is greater than 0.5 torr for 30 minutes. A fault will occur if the pressure is greater than 0.5 torr for 5 minutes after the warning is triggered.	• Verify the foreline valve is opening properly. • Verify there is control air hooked up to the system. • If in manual mode, verify the foreline valve wasn't closed too long.
26	Chiller Response	The chiller doesn't appear to be responding: • Not receiving setpoints from 3500 • Not activating its thermal control when its supposed to • Not de-activating its thermal control when its supposed to	• The likely reason for any of these events is the chiller is in some kind of a fault state. • Check the chiller manual to troubleshoot the fault. • Power cycle the chiller
27	Watlow Comms	Communication with the watlow controller has been lost	• Contact Laco Technologies for support.
28	TC Card Comms	Communication with the Thermal Couple Card was lost	• Contact Laco Technologies for support.

8. Maintenance and Spare Parts

The recommended spare for the 3500 controller are summarized in the table below. For maintenance and spare parts for items located outside of the 3500 controller refer to the 3500 system manual for more information

NOTICE

Take note of all parts noted as critical spares and all recommended replacement intervals in this section

Items identified as critical spares typically have a high likelihood of needing replacement so the 3500 Controller can stay in proper operating condition. Also, critical spares include parts that if damaged or worn may have long lead times to get replacements. Part listed that are not critical spares may fail, but are more unlikely than critical spares. Take special note of the replacement intervals in the tables below as it is always best to replace a component before it fails and leads to equipment downtime.

Laco Technologies offers a variety of service agreements so that companies can have assistance, if they desire, in maintaining their 3500 system. If any maintenance or service agreement questions arise [contact Laco Technologies](#).

3500 Controller Spare Parts and Maintenance List				
ID	Part Number	Description	Critical Spare	Replacement Interval
1	LMSA3849	FUSE, 5X20MM, 1 AMP, SLOW BLOW, GLASS		As needed
2	LMSA3851	FUSE, 5X20MM, 3 AMP, MEDIUM TIME DELAY, GLASS		As needed
3	LMSA5133	RELAY, SOCKET, FOR 782 DPDT SERIES RELAYS, USE RELAY		As needed
4	LMSA5422	Touchscreen, 7"		As needed

9. Appendix

9.1. Communications

The VC/HVC 3500 controller has the ability to communicate with other devices using ASCII code through ethernet TCP and RS232.

The RS232 settings are:

- Baud Rate: 9600
- Data Bit: 8
- Parity List: None
- Stop Bits: 1

For Ethernet TCP ensure the correct IP address has been entered. For the VC/HVC 3500 controllers there are two IP address, one for the panel, the other for the CPU. For communications the CPU IP address is used.

9.1.1. Commands and Examples

- All write commands start with “!” (ASCII 33)
- All read commands start with “?” (ASCII 63)
- All send and reply commands end with carriage return (0x13)
- If the PLC doesn’t recognize a command, it will respond with “ER[CR]”

Command	Description	Response
!CS	Start Cycle	CS
!CA	Abort Cycle	CA
?TC	Test Status	(status)
?MC	System Mode	AUTO or MANUAL
!TR:##	Change Recipe	TR:##
?TR	Current Recipe	TR:##
?RS	Current Recipe Step	RS:##
!RS	Recipe Start	RS
!RO	Recipe Stop	RO
?VS	Vacuum Setpoint	VS:####.#####
!VS:#####	Change Vac Setpoint	VS:####.#####
?VR	Vac Cntrl Range	VR:##.###
!VR:#####	Set Vac Cntrl Range	VR:##.###
!VD:#####	Set Vac Rate Cntrl	VD:##.###
?VD	Vac Rate Cntrl	VD:##.###
?VH	Hold Time	VH:#####
!VH:#####	Set Hold Time	VH:#####
?TT	Test Time	TT:#####
?VP	Vacuum Pressure	VP:####.#####
?PR	Pressure Rate	PR:#####
!FN:#####	Fan RPM	FN:#####
?T## (see notes)	Ask Temperature	T##:####
?Z# (see notes)	Ask Temp. Setpoint	Z#:####
!Z#:##### (see notes)	Change Temp Setpoint	Z#:####

?ZR# (see notes)	Ask Temp Rate Setpoint	ZR#####
!ZR##### (see notes)	Change Rate Setpoint	ZR#####
?RT# (see notes)	Ask Temp cntrl Range	RT#####
!RT##### (see notes)	Change Temp cntrl Range	RT#####
!ZS# (see notes)	Activate Thermal Cntrl	ZS#
!ZO# (see notes)	Deactivate Thermal Cntrl	ZO#
!VA	Vent to Atmosphere	VA
!FA	Fill to Atmosphere	FA
!PS	Purge System	PS
!NA	None (close all valves)	NA
?ES	Get the Error Status	ES: {S}: ##### *
!CR	Resets the system	CR
!OR	Toggles Rough Valve	OR: {S}**
?OR	Status of Rough Valve	OR: {S}
!OV	Toggles Vent Valve	OV: {S}
?OV	Status of Vent Valve	OV: {S}
!OF	Toggles Fill Valve	OF: {S}
?OF	Status of Fill Valve	OF: {S}
!O4	Toggles Foreline Valve	O4: {S}
?O4	Status of Foreline Valve	O4: {S}
!OG	Toggles Gate Valve	OG: {S}
?OG	Status of Gate Valve	OG: {S}
!OP	Toggles Pump	OP: {S}
?OP	Status of Pump	OP: {S}
!OT	Toggles Turbo Pump	OT: {S}
?OT	Status of Turbo Pump	OT: {S}

*The error code will return two values:

- ES: standard return indicating the command that was sent)
- S: this is representative of three possible letters:
 - N: No warnings or faults
 - W: Warning Active
 - F: Fault Active
- #####: This is a UINT32 integer number that represents which warning/fault is active. See warnings/faults numbers table in section 7. You will need to convert the UINT32 number into its 32 bits. The bit position represents the warning/fault number. A 0 = not active, a 1 = active.
- Example: The air compressor faulted and the shop no longer has any compressed air. Warning number 16 (the compressed air warning) on the TVAC becomes activated. The 32 bit binary number that represents this scenario is: 00000000000000010000000000000000. Notice the 16th position in this number is high, this represents the status of warning number 16. This number converted to a UINT32 decimal is 65536. So if the user sends the code “?ES{CR}”. The controller will respond with “ES: W: 000065536”
- Example 2: Someone accidentally unplugs the ethernet cablet that connects to the high vacuum box and the chiller. The system faults and activates the “Chiller

Comms” warning, and the “HV Comms” warning. These are warnings numbers 20 and 22 respectively. The 32 bit binary representation of this is: 00000000010100000000000000000000. Notice out positions 20 and 22 are high. The UINT32 decimal representation of this number is: 4194304. So when the user sends the code “?ES{CR}”. The controller will respond with “ES: F: 00004194304{CR}”

** The status of a valve or pump will either be “C” for closed or “O” for open. For example: if the user sends the command “?OR{CR}”. If the rough valve is open, the controller will respond with “OR: O{CR}”.

- Please note the status of the valve is given at the time the code is sent. If the rough valve is close and the user sends “!OR{CR}”. The response will be “OR: C{CR}”. Which indicates the valve is closed. Which is true. The command will open the valve about 1 second after the status response is sent.

Notes and examples:

- Temperature
- All temperature read and writes do not have a decimal (they are integers). To write the value of 56.8, type it with the decimal one spot to the right: 568. Conversely, if the PLC writes the following temperature: 754, you need to move the decimal to the left one spot: 75.4.
 - The PLC has storage for 21 different temperature sensors. The ascii code will include which one of those 21 temperature storages you want to read.
 - Example: On a system that has a RTD on a shelf, and another RTD on the wall where the heater is. The shelf RTD is being read by temp0 and the wall RTD is being read by temp3, the user would send the following command:
 - ?T0[CR], to which the PLC will respond: T0:458[CR]
 - ?T3[CR], to which the PLC will respond: T3:684[CR]
 - Since the software is designed to have three zones, the ascii code needs to designate which zone the code is being applied to. Remember, the majority of systems only have one zone.
 - Example, the user would like to change the temperature setpoint for zone 1 to 85.5 degrees, the code would be:
 - !Z1:855[CR], to which the PLC will respond: Z1:855[CR]
 - Example, the user would like to ask what the temperature rate setpoint for zone 2 is:
 - ?ZR2, to which the PLC will respond: ZR2:4[CR] (in this example the temperature rate setpoint is 0.4°C/min).
 - Example, the user would like to change the temperature control range for zone 1 to 1.5:
 - !RT1:15[CR], to which the PLC will respond: RT1:15[CR]
 - Example, the user wants to activate the thermal control on zone 2, the code to send is:
 - !ZS2, to which the PLC will respond: ZS2

Setting up the controller to run a cycle from ASCII codes:

Although you can send a code at any time during a cycle or manual mode, you may want to just program a cycle on your computer software and then have that run the program. To do this go to your recipe screen and select ASCII as your vacuum operation (ensure the “Enable ASCII in Recipe” feature is turned on in the advanced settings. Set the trigger to hold. It will look like this:

Recipe - Vacuum & Thermal						
Recipe: 2 - Water Mark				Last Step Number: 0		
Step #	Vacuum Step	Vac SetPoint Torr	Hold Time	Hold Step	Optional Step	Zone1
0	ASCII	0.00	00:00:00	☒	Off	0.0
1	Setpoint	0.00	00:00:00	☐	Off	0.0
2	Setpoint	0.00	00:00:00	☐	Off	0.0
3	Setpoint	0.00	00:00:00	☐	Off	0.0
4	Setpoint	0.00	00:00:00	☐	Off	0.0

↑
↓
Thermal Settings
🏠

You can leave the setpoint and thermal settings turned off.

Go to the run screen. If you press start (or send the command !CS[CR]) the PLC will wait for commands coming from the computer.

Example:

- A cycle needs to be carried out to pump the chamber down to 10 torr and hold that pressure to within +/-2 torr. (note the pressure units will be whatever is set up in the advanced settings).
- Once it reaches 10 torr, turn on the heater and heat the shelf to 80.5°C and hold that temperature to +/-2 degrees for 30 minutes
- While keeping the heat on the system, pressurize the chamber to 1000 torr. Hold this pressure and temperature for an hour.
- Turn off heater and vent to atmosphere
- During the process you want to sample the pressure and temperature every 20 seconds

The pseudo code would look like the following:

- !CS[CR] (starts the cycle)
- ?VP[CR] (reads the chamber pressure) set this code to repeat every 20 seconds
- ?T1[CR] (reads the shelf temperature) set this code to repeat every 20 seconds
- !VR:2[CR] (sets the vacuum control range to 2 torr)
- !VS:10[CR] (sets the vacuum set point to 10 torr).
 - At this point the system will open the evac valve and the chamber will begin to pump down.
- Wait until the system reached 10 torr
- !ZS (activate thermal control for zone 1)
- !Z1:805[CR] (set the temperature setpoint to 80.5)
- Wait for the temperature to reach setpoint.
- Once it reaches the setpoint, wait for 30 minutes, the PLC will maintain 80.5°C
- !VS:1000[CR] (set the pressure to 1000 torr).
- Wait until the chamber reaches 1000 torr
- Wait for an hour
- !CS [CR](simulates the start button). Since the trigger is set to hold, and all parameters are within their setpoints, the PLC is just waiting for the start button to be pressed to return the chamber to ambient conditions. By pressing the start button (or !CS[CR]) the chamber will vent to turn off the heater.